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MITgcm.jl: a Julia Interface to the MITgcm

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Summary

General circulation models are used to study climate, ocean and atmosphere dynamics, biogechemistry, ecology, and more. The `MITgcm`, written in `Fortran`, is one of the most widely-used models of this class. We present `MITgcm.jl`, a two-way interface to `MITgcm` written in Julia that can be used to not only analyze model results but also drive model simulations. With `MITgcm.jl` users can setup, build, and launch `MITgcm` without having to know shell scripting or having to edit text files manually. `MITgcm.jl` provides support in `Julia` for the various input and output formats used in `MITgcm`. It implements the `ClimateModels.jl` interface, and opens up a whole new way of using `MITgcm` interactively from Jupyter and Pluto notebooks. `MITgcm.jl` in turn brings full-featured, reliable ocean modeling to Julia.

Statement of Need

The cutting-edge of climate modeling, and much of its legacy, is based on numerical models written in compiled languages like `Fortran` or `C`. The MIT general circulation model (`MITgcm`) for example now runs globally at the kilometer scale to provide an unprecedented view of ocean dynamics (Fig.~1, Campin et al. (2004), Marshall et al. (1997), Gallmeier et al. (2023)). With its unrivaled adjoint modeling capabilities, `MITgcm` is also the computational engine that powers the ECCO ocean reanalysis, a widely-used data-assimilating product in climate science (Fig. 1, Heimbach et al. (2002), G. Forget et al. (2015), Gaël Forget (2024a)). `MITgcm` additionally provides unique modeling capabilities for ocean biogeochemistry, ecology, and optics (Dutkiewicz et al. (2015), Gaël Forget (2019)). While a new generation of models, written in languages like C++ and Julia, is poised to better exploit GPUs (Ramadhan et al. (2020), Wu & Forget (2022), E3SM Project (2024)), `Fortran`-based models are expected to remain popular on other computer architectures for the foreseeable future. They also provide a crucial reference point to evaluate next generation models.

Models like `MITgcm` benefit from decades of accumulated expertise, careful inspection, and continuous integration. Running such a model or exploiting its results does not require knowing the computer language being used internally. However, `Fortran`-based models can sometime appear complicated or inconvenient to operate due to technical hurdles like having to use a compiler directly, to edit text files manually, or to deal with shell-scripting. Fortunately, such issues are easily alleviated by providing a user-friendly interface (e.g., Fig. 2) written in a high-level language (e.g., `Julia`) to interact with climate models written in lower-level languages as done here for `MITgcm`.

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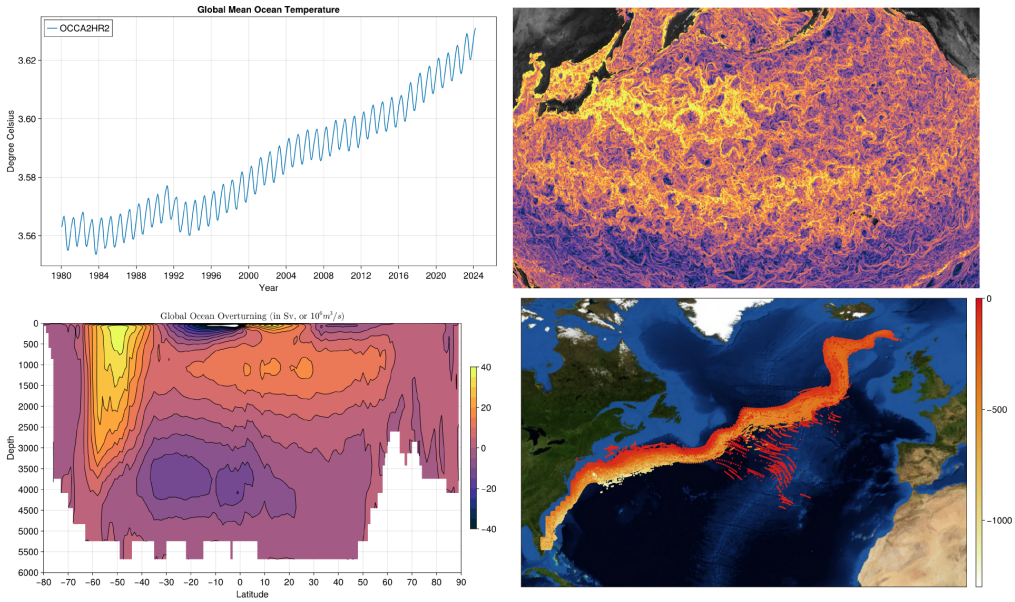

Figure 1: Examples of `MITgcm` output created, read, processed, plotted, and analyzed via `MITgcm.jl` in `Julia`. Top left : global mean ocean warming over 1980-2023 as simulated in `OCCA2HR2` using `MITgcm.jl` (Gaël Forget, 2024a). Bottom Left : climatological mean view of the global ocean conveyor belt (Rousselet et al., 2021) produced in `Climatology.jl`. Bottom right : particle tracking of seawater pathtways along the Gulf Stream, computed in three dimensions by `IndividualDisplacements.jl` (Gaël Forget, 2021). Top Right : temperature fronts in a global `MITgcm` simulation, which ran on a 4km resolution Lat-Lon-Cap grid (G. Forget et al., 2015, p. Gallmeier2023), diagnozed and visualized in `Julia` using `MeshArrays.jl`.

`MITgcm.jl` can read the various types of output that `MITgcm` generates. This feature enables not only common analyses of model results (e.g., mapping and plotting), but also accurate computations of quantities such as ocean heat transport and global warming (Fig. 1, G. Forget et al. (2015), Gaël Forget & Ferreira (2019), Gaël Forget (2024a)). `MITgcm.jl` also makes it easy to deploy and run any configuration of `MITgcm` on laptops, HPC clusters, and in the cloud. `MITgcm.jl` interacts with `MITgcm`'s run-time model parameters in `Julia` via ordered dictionaries. It can re-export them to the standard `TOML` file format, or to the native `MITgcm` format. Owing to this two-way interface, `MITgcm` can now be used from Jupyter or Pluto notebooks interactively via `MITgcm.jl`. The package includes a series of examples and tutorials that demonstrate the interface (e.g., Fig. 2).

In the code example below, `MITgcm_config` defines the `MC` data structure. The `run` command is equivalent to the sequence of `setup(MC); build(MC); launch(MC)`. And `readdir` returns a list of files found in the folder where `MITgcm` ran (a temporary folder by default).

```
using MITgcm
MC=MITgcm_config(configuration="tutorial_held_suarez_cs")
run(MC)
readdir(MC,"run")
```

We can then modify parameters in julia, call `write_all_namelists` to update the `MITgcm` run-time parameter files accordingly, and rerun a new model simulation in the same folder. In the example below, we thus extend the simulation to 64 time steps.

```
MC.inputs[:main][:PARM03][:nTimeSteps]=64
write_all_namelists(MC.inputs,joinpath(MC,"run"))
launch(MC)
```

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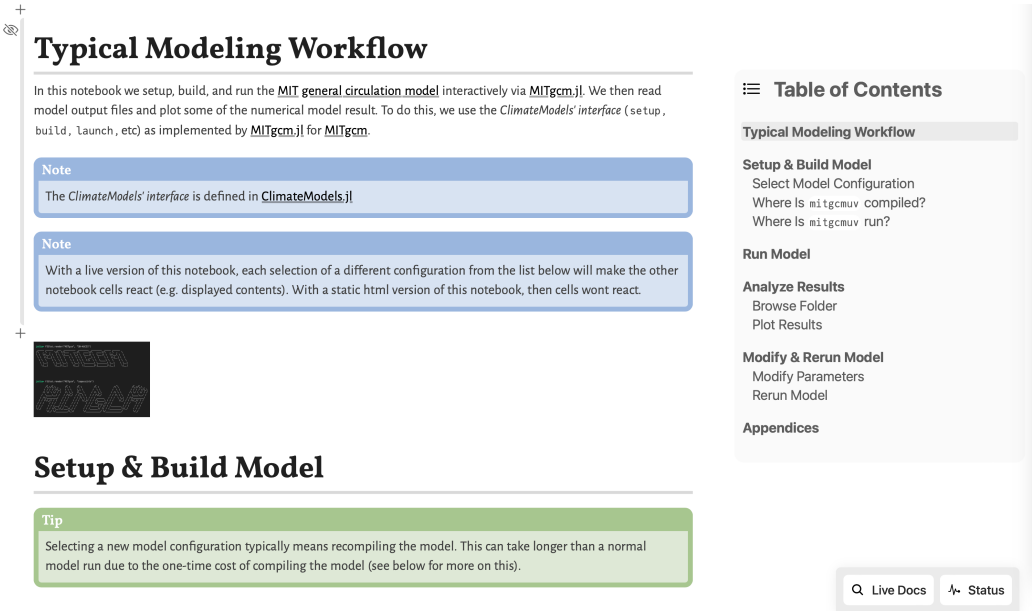

Figure 2: Notebook that operates `MITgcm` interactively, and let's user visualize model results without having to write code. Both Jupyter and Pluto notebooks are supported.

`MITgcm.jl` brings all of `MITgcm`'s modeling capabilities to a new category of users, who may not be trained in `Fortran` or shell-scripting. Furthermore, `MITgcm.jl` implements the `ClimateModels.jl` interface (Gaël Forget, 2024b), which (1) streamlines the handling of file folders, (2) makes it easier to run and rerun simulations, and (3) supports an extensive lineup of complementary models written in various languages. `MITgcm.jl` can also be used to build integrated cyberinfrastructure solutions as demonstrated in Duckworth et al. (2023).

`MITgcm.jl` in turn provides the vast `MITgcm` user community with a bridge to new tools for machine learning, artificial intelligence, differential equations, visualization, etc from the `Julia` software stack. Examples include the use of `MITgcm` output in offline mode to estimate sea water pathways and ocean transports (Fig. 1, Gaël Forget (2021)), or simulate the behavior of marine ecosystems using an agent-based modeling approach (Wu & Forget, 2022).

Video Presentations

Video presentations listed below further demonstrate key features of `MITgcm.jl`.

- MITgcm demo that uses the `ClimateModels.jl` interface to run `MITgcm` via `MITgcm.jl`.
- JuliaCon2021 `ClimateModels.jl` was presented at JuliaCon in 2021.
- JuliaCon2023 `ClimateModels.jl` and `MITgcm.jl` were further presented at JuliaCon in 2023.

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